

bradley_spring22: formula evidence

The page, the confidence and the picture of an inline formula are its HOST LINE’s — a formula has none of its own. A line’s confidence is not a formula’s.

Inline formulas (first occurrence)

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bradley_spring22_F00001	3	■ 1.000	p	p	relationship as a simple fraction. If an event occurs with probability p , then we might say the
bradley_spring22_F00002	3	■ 1.000	1 / p	$1/p$	amount of information conveyed is $1/p$ because if p is small, then $1/p$ is large and vice versa.
bradley_spring22_F00003	3	■ 1.000	p=1	$p = 1$	$p = 1$. Since the event is certain to occur, it would not be surprising to learn that it did
bradley_spring22_F00004	3	■ 1.000	1 / p=1 / 1=1	$1/p = 1/1 = 1$	$1/p = 1/1 = 1$ is not zero, which goes against that intuition. On the other hand, if $p = 1$, then
bradley_spring22_F00005	3	■ 1.000	\log (1 / p)=\log (1)=0	$\log(1/p) = \log(1) = 0$	$\log(1/p) = \log(1) = 0$ as desired. This is one reason why logarithms are more convenient.
bradley_spring22_F00006	3	■ 1.000	\log (1 / p)	$\log(1/p)$	with probability p to be the number $\log(1/p)$.
bradley_spring22_F00007	3	■ 1.000	x	x	¹ The logarithm of some number x is another number that we will denote by $\log(x)$ (taken to be the natural logarithm
bradley_spring22_F00008	3	■ 1.000	\log (x)	$\log(x)$	¹ The logarithm of some number x is another number that we will denote by $\log(x)$ (taken to be the natural logarithm
bradley_spring22_F00009	3	■ 1.000	\log (1)=0	$\log(1) = 0$	in this paper). In particular, a useful fact to know in this article is that $\log(1) = 0$.
bradley_spring22_F00010	6	■ 1.000	2 \times 3=6	$2 \times 3 = 6$	combine them, say by multiplication, to form a new number: $2 \times 3 = 6$. Of course there is
bradley_spring22_F00011	6	■ 1.000	y	y	bers, say x and y , then we can multiply them to obtain a new number $x \times y$, also denoted by
bradley_spring22_F00012	6	■ 1.000	x \times y	$x \times y$	bers, say x and y , then we can multiply them to obtain a new number $x \times y$, also denoted by
bradley_spring22_F00013	7	■ 1.000	x y	xy	xy . This simple idea of combining things (whether by multiplication or addition or something
bradley_spring22_F00014	7	■ 1.000	\left(x^{\mathrm{3}}\right)^{\mathrm{2}} y^{\mathrm{4}} x^{-1}	$\left(x^3\right)^2 y^4 x^{-1}$	assignments such as “Simplify the expression $(x^3)^2 y^4 x^{-1}$ using laws of exponents” and “Fac-
bradley_spring22_F00015	7	■ 1.000	x^{\mathrm{2}}+7 x+12^{\prime \prime}	$x^2+7 x+12''$	tor the quadratic polynomial $x^2+7 x+12'$ and the like. But this kind of algebra is nothing
bradley_spring22_F00016	7	■ 1.000	(2 \times 3) \times 5	$(2 \times 3) \times 5$	plied first: $(2 \times 3) \times 5$ is the same as $2 \times (3 \times 5)$, and the analogous holds for concatenation of
bradley_spring22_F00017	7	■ 1.000	2 \times(3 \times 5)	$2 \times (3 \times 5)$	plied first: $(2 \times 3) \times 5$ is the same as $2 \times (3 \times 5)$, and the analogous holds for concatenation of
bradley_spring22_F00018	7	■ 0.985	A_{\infty}	A_∞	include commutative algebras, associative algebras, Lie algebras, A_∞ -algebras, and more. ² In
bradley_spring22_F00019	8	■ 1.000	20=19.999 \ldots	$20 = 19.999 \dots$	as decimals, and then have to mention aliases like $20 = 19.999 \dots$, a fact not unknown to the
bradley_spring22_F00020	9	■ 0.982	X^{\prime \prime}	X''	Very roughly speaking, “a topology on a set X ” is a mathematician’s way of declaring which
bradley_spring22_F00021	9	■ 0.983	X	X	elements in X are close to each other. When considered together as a pair, both the set and its
bradley_spring22_F00022	10	■ 0.803	\left(\frac{1}{2}, \frac{1}{2}\right)	$\left(\frac{1}{2}, \frac{1}{2}\right)$	whose sum is 1. For example, $\left(\frac{1}{2}, \frac{1}{2}\right)$ and $\left(\frac{2}{5}, \frac{1}{2}, \frac{1}{10}\right)$ are both probability distributions whereas
bradley_spring22_F00023	10	■ 0.803	\left(\frac{2}{5}, \frac{1}{2}, \frac{1}{10}\right)	$\left(\frac{2}{5}, \frac{1}{2}, \frac{1}{10}\right)$	whose sum is 1. For example, $\left(\frac{1}{2}, \frac{1}{2}\right)$ and $\left(\frac{2}{5}, \frac{1}{2}, \frac{1}{10}\right)$ are both probability distributions whereas
bradley_spring22_F00024	10	■ 1.000	(7,3)	$(7,3)$	$(7,3)$ and $\left(-\frac{2}{5}, -\frac{1}{2}, -\frac{1}{10}\right)$ are not. In the first example, $\frac{1}{2}$ and $\frac{1}{2}$ may represent the respective

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bradley_spring22_F00025	10	■ 1.000	$\left(-\frac{2}{5}, -\frac{1}{2}, -\frac{1}{10}\right)$	$\left(-\frac{2}{5}, -\frac{1}{2}, -\frac{1}{10}\right)$	$(7,3)$ and $\left(-\frac{2}{5}, -\frac{1}{2}, -\frac{1}{10}\right)$ are not. In the first example, $\frac{1}{2}$ and $\frac{1}{2}$ may represent the respective
bradley_spring22_F00026	10	■ 1.000	$\frac{1}{2}$	$\frac{1}{2}$	$(7,3)$ and $\left(-\frac{2}{5}, -\frac{1}{2}, -\frac{1}{10}\right)$ are not. In the first example, $\frac{1}{2}$ and $\frac{1}{2}$ may represent the respective
bradley_spring22_F00027	10	■ 1.000	$\frac{2}{5}, \frac{1}{2}$	$\frac{2}{5}, \frac{1}{2}$	probabilities of landing a heads or tails on a fair coin toss, while in the second example $\frac{2}{5}, \frac{1}{2}$,
bradley_spring22_F00028	10	■ 1.000	$\frac{1}{10}$	$\frac{1}{10}$	and $\frac{1}{10}$ may represent the respective probabilities of choosing cereal, oatmeal, or fruit for
bradley_spring22_F00029	10	■ 0.960	n	n	Said more formally, given a natural number n (that is, a whole number $1, 2, 3, \dots$), a proba-
bradley_spring22_F00030	10	■ 0.997	$\{1, 2, \dots, n\}$	$\{1, 2, \dots, n\}$	bility distribution on a set $\{1, 2, \dots, n\}$ is a list of nonnegative real numbers $p = (p_1, p_2, \dots, p_n)$
bradley_spring22_F00031	10	■ 0.997	$p = \left(p_1, p_2, \dots, p_n\right)$	$p = (p_1, p_2, \dots, p_n)$	bility distribution on a set $\{1, 2, \dots, n\}$ is a list of nonnegative real numbers $p = (p_1, p_2, \dots, p_n)$
bradley_spring22_F00032	11	■ 1.000	$p_1 + p_2 + \dots + p_n = 1$	$p_1 + p_2 + \dots + p_n = 1$	satisfying $p_1 + p_2 + \dots + p_n = 1$. Elements in the set $\{1, 2, \dots, n\}$ may be thought of as enu-
bradley_spring22_F00033	11	■ 0.917	$\left(p_1, p_2, \dots, p_n\right)$	$(p_1, p_2, \dots, p_n)$	The letter p is being used as shorthand to represent the full list of numbers $(p_1, p_2, \dots, p_n)$.
bradley_spring22_F00034	11	■ 1.000	$n=2$	$n = 2$	Teasing this out with $n = 2$, suppose we have a set of two elements $\{1, 2\}$ corresponding to
bradley_spring22_F00035	11	■ 1.000	$\{1, 2\}$	$\{1, 2\}$	Teasing this out with $n = 2$, suppose we have a set of two elements $\{1, 2\}$ corresponding to
bradley_spring22_F00036	11	■ 1.000	$p_1 = \frac{1}{2}$	$p_1 = \frac{1}{2}$	the two faces of a coin, namely heads (option #1) or tails (option #2). If we let $p_1 = \frac{1}{2}$ and
bradley_spring22_F00037	11	■ 1.000	$p_2 = \frac{1}{2}$	$p_2 = \frac{1}{2}$	$p_2 = \frac{1}{2}$, then the list $p = (p_1, p_2)$ is the first example of a probability distribution given above.
bradley_spring22_F00038	11	■ 1.000	$p = \left(p_1, p_2\right)$	$p = (p_1, p_2)$	$p_2 = \frac{1}{2}$, then the list $p = (p_1, p_2)$ is the first example of a probability distribution given above.
bradley_spring22_F00039	11	■ 1.000	$p = \left(\frac{1}{2}, \frac{1}{2}\right)$	$p = \left(\frac{1}{2}, \frac{1}{2}\right)$	pictures. Figure 2 gives an example. There, the probability distribution $p = \left(\frac{1}{2}, \frac{1}{2}\right)$ associated
bradley_spring22_F00040	11	■ 1.000	$H(p)$	$H(p)$	This number, which we will denote by $H(p)$, is given explicitly in the following formula.
bradley_spring22_F00041	12	■ 0.966	$\log(1) - \log(p)$	$\log(1) - \log(p)$	of logarithms, this number is the same as $\log(1) - \log(p)$ which is equal to $-\log(p)$ because
bradley_spring22_F00042	12	■ 0.966	$-\log(p)$	$-\log(p)$	of logarithms, this number is the same as $\log(1) - \log(p)$ which is equal to $-\log(p)$ because
bradley_spring22_F00043	12	■ 1.000	$-\log\left(p_1\right)$	$-\log(p_1)$	first computes the information associated to each event, that is, $-\log(p_1)$ and $-\log(p_2)$ and
bradley_spring22_F00044	12	■ 1.000	$-\log\left(p_2\right)$	$-\log(p_2)$	first computes the information associated to each event, that is, $-\log(p_1)$ and $-\log(p_2)$ and
bradley_spring22_F00045	12	■ 0.983	$\operatorname{logarithm}\log(p)$	$\operatorname{logarithm}\log(p)$	are important. The natural logarithm $\log(p)$ is negative whenever its input p is between 0 and
bradley_spring22_F00046	12	■ 1.000	$p = (1, 0)$	$p = (1, 0)$	to the probability distribution $p = (1, 0)$. If we were to learn that this person indeed drank
bradley_spring22_F00047	12	■ 1.000	$\frac{1}{n}$	$\frac{1}{n}$	comes each with equal probability $\frac{1}{n}$, the entropy of the resulting probability distribution
bradley_spring22_F00048	12	■ 1.000	$\left(\frac{1}{n}, \frac{1}{n}, \dots, \frac{1}{n}\right)$	$\left(\frac{1}{n}, \frac{1}{n}, \dots, \frac{1}{n}\right)$	$\left(\frac{1}{n}, \frac{1}{n}, \dots, \frac{1}{n}\right)$ will always be $\log(n)$, and this turns out to be the maximum possible value. In
bradley_spring22_F00049	12	■ 1.000	$\log(n)$	$\log(n)$	$\left(\frac{1}{n}, \frac{1}{n}, \dots, \frac{1}{n}\right)$ will always be $\log(n)$, and this turns out to be the maximum possible value. In
bradley_spring22_F00050	12	■ 1.000	$0 \leq H(p) \leq \log(n)$	$0 \leq H(p) \leq \log(n)$	other words, one can show that $0 \leq H(p) \leq \log(n)$ for any probability distribution p on a set

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bradley_spring22_F00051	13	■ 1.000	H	H	We have used the letter H as the name of the function that assigns to a probability distribution
bradley_spring22_F00052	13	■ 1.000	$\Delta_{\{n\}}$	Δ_n	p its entropy $H(p)$. We will also use Δ_n to denote the set of all possible probability distributions
bradley_spring22_F00053	13	■ 1.000	$\Delta^{\{n-1\}}$	Δ^{n-1}	and so on. Mathematicians usually prefer to lower the index by 1 and write Δ^{n-1} instead,
bradley_spring22_F00054	13	■ 1.000	$\mathbb{R}$	$\mathbb{R}$	Δ_n instead. ³ In our menagerie of mathematical notation, we have also used the symbol $\mathbb{R}$ to
bradley_spring22_F00055	13	■ 1.000	$H: \Delta_{\{n\}} \rightarrow \mathbb{R}$	$H: \Delta_n \rightarrow \mathbb{R}$	the assignment of a probability distribution p to the number $H(p)$. The notation $H: \Delta_n \rightarrow \mathbb{R}$
bradley_spring22_F00056	13	■ 1.000	$f: X \rightarrow Y$	$f: X \rightarrow Y$	is very convenient, so we will always use analogous notation $f: X \rightarrow Y$ to mean “a function f
bradley_spring22_F00057	13	■ 1.000	f	f	is very convenient, so we will always use analogous notation $f: X \rightarrow Y$ to mean “a function f
bradley_spring22_F00058	13	■ 1.000	Y	Y	from a set X to a set Y that assigns to each input x in X one output $f(x)$ in Y .”
bradley_spring22_F00059	13	■ 1.000	$f(x)$	$f(x)$	from a set X to a set Y that assigns to each input x in X one output $f(x)$ in Y .”
bradley_spring22_F00060	14	■ 0.979	$n=1,2,3, \ldots$	$n = 1, 2, 3, \dots$	are infinitely many natural numbers $\overline{i = 1, 2, 3, \dots}$, there are necessarily infinitely many H s,
bradley_spring22_F00061	14	■ 1.000	$H_{\{n\}}$	H_n	as well. It would be helpful to indicate this dependency in our notation and write H_n instead
bradley_spring22_F00062	14	■ 1.000	$H: \Delta_{\{1\}} \rightarrow \mathbb{R}$	$H: \Delta_1 \rightarrow \mathbb{R}$	$H: \Delta_1 \rightarrow \mathbb{R}$ and $H: \Delta_2 \rightarrow \mathbb{R}$ and $H: \Delta_3 \rightarrow \mathbb{R}$, and so on.
bradley_spring22_F00063	14	■ 1.000	$H: \Delta_{\{2\}} \rightarrow \mathbb{R}$	$H: \Delta_2 \rightarrow \mathbb{R}$	$H: \Delta_1 \rightarrow \mathbb{R}$ and $H: \Delta_2 \rightarrow \mathbb{R}$ and $H: \Delta_3 \rightarrow \mathbb{R}$, and so on.
bradley_spring22_F00064	14	■ 1.000	$H: \Delta_{\{3\}} \rightarrow \mathbb{R}$	$H: \Delta_3 \rightarrow \mathbb{R}$	$H: \Delta_1 \rightarrow \mathbb{R}$ and $H: \Delta_2 \rightarrow \mathbb{R}$ and $H: \Delta_3 \rightarrow \mathbb{R}$, and so on.
bradley_spring22_F00065	14	■ 1.000	$\Delta_{\{1\}}, \Delta_{\{2\}}, \Delta_{\{3\}}, \ldots$	$\Delta_1, \Delta_2, \Delta_3, \dots$	portantly, each of these sets $\overline{\Delta_1, \Delta_2, \Delta_3, \dots}$ is not merely a set. There is a standard way in
bradley_spring22_F00066	14	■ 1.000	$\Delta_{\{1\}}$	Δ_1	be shown that Δ_1 is a point, Δ_2 is a line segment, Δ_3 is a triangle, and Δ_4 is a pyramid, as
bradley_spring22_F00067	14	■ 1.000	$\Delta_{\{2\}}$	Δ_2	be shown that Δ_1 is a point, Δ_2 is a line segment, Δ_3 is a triangle, and Δ_4 is a pyramid, as
bradley_spring22_F00068	14	■ 1.000	$\Delta_{\{3\}}$	Δ_3	be shown that Δ_1 is a point, Δ_2 is a line segment, Δ_3 is a triangle, and Δ_4 is a pyramid, as
bradley_spring22_F00069	14	■ 1.000	$\Delta_{\{4\}}$	Δ_4	be shown that Δ_1 is a point, Δ_2 is a line segment, Δ_3 is a triangle, and Δ_4 is a pyramid, as
bradley_spring22_F00070	14	■ 1.000	$n=1$	$n = 1$	► <i>In more detail.</i> Consider the case when $n = 1$. The set Δ_1 consists of all probability dis-
bradley_spring22_F00071	15	■ 0.985	x, y	x, y	numbers (x, y) whose sum is 1; that is, $x + y = 1$. We can rearrange this equation to see it
bradley_spring22_F00072	15	■ 0.985	$x+y=1$	$x + y = 1$	numbers (x, y) whose sum is 1; that is, $x + y = 1$. We can rearrange this equation to see it
bradley_spring22_F00073	15	■ 1.000	$y=1-x$	$y = 1 - x$	is equivalent to $y = 1 - x$, which is the equation of a line. Since we have further restricted x
bradley_spring22_F00074	15	■ 1.000	$n=3$	$n = 3$	distribution on two elements. Moving on to the case when $n = 3$, the set Δ_3 of probability
bradley_spring22_F00075	15	■ 0.999	x, y, z	x, y, z	distributions on three elements is the set of all triples of nonnegative numbers (x, y, z) whose
bradley_spring22_F00076	15	■ 0.922	$x+y+z=1$	$x + y + z = 1$	sum is 1; that is $x + y + z = 1$ or equivalently $z = 1 - x - y$, which is the equation of a plane
bradley_spring22_F00077	15	■ 0.922	$z=1-x-y$	$z = 1 - x - y$	sum is 1; that is $x + y + z = 1$ or equivalently $z = 1 - x - y$, which is the equation of a plane
bradley_spring22_F00078	15	■ 1.000	$n=4$	$n = 4$	case when $n = 4$ requires more work, but it can be shown that the set Δ_4 of all probability

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bradley_spring22_F00079	15	■ 1.000	n-1	$n - 1$	the topological space Δ_n is called an $n - 1$ - simplex . ⁴ For small values of n we have seen that
bradley_spring22_F00080	17	■ 0.516	10 \%	10%	oatmeal, and a 10% chance we choose fruit. This probability distribution on three breakfast
bradley_spring22_F00081	17	■ 1.000	$q=\left(\frac{2}{5}, \frac{1}{2}, \frac{1}{10}\right)$	$q = \left(\frac{2}{5}, \frac{1}{2}, \frac{1}{10}\right)$	items will be denoted by $q = \left(\frac{2}{5}, \frac{1}{2}, \frac{1}{10}\right)$, which is a point in Δ_3 . The picture for this probability
bradley_spring22_F00082	17	■ 1.000	$r=\left(\frac{3}{10}, \frac{7}{10}\right)$	$r = \left(\frac{3}{10}, \frac{7}{10}\right)$	$r = (\frac{3}{10}, \frac{7}{10})$, which is a point in Δ_2 . The picture of this probability distribution is shown on
bradley_spring22_F00083	17	■ 1.000	q	q	tribution q on three elements
bradley_spring22_F00084	17	■ 1.000	r	r	larly for r .
bradley_spring22_F00085	17	■ 0.977	$\frac{1}{2} \times \frac{2}{5} = \frac{1}{5}$	$\frac{1}{2} \times \frac{2}{5} = \frac{1}{5}$	product of the probabilities of each individual outcome, namely $\frac{1}{2} \times \frac{2}{5} = \frac{1}{5}$ or 20%. Similarly,
bradley_spring22_F00086	17	■ 1.000	$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$	$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$	equal to $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$ or 25%, and the probability of flipping heads and choosing fruit is equal
bradley_spring22_F00087	17	■ 0.988	$\frac{1}{2} \times \frac{1}{10} = \frac{1}{20}$	$\frac{1}{2} \times \frac{1}{10} = \frac{1}{20}$	to $\frac{1}{2} \times \frac{1}{10} = \frac{1}{20}$ or 5%. This, too, has a visual counterpart. One can imagine grafting the root of
bradley_spring22_F00088	17	■ 1.000	$\frac{3}{20}$	$\frac{3}{20}$	flipping tails and choosing pizza is $\frac{3}{20}$, and the probability of flipping tails and choosing stir
bradley_spring22_F00089	17	■ 1.000	$\frac{7}{20}$	$\frac{7}{20}$	fry is $\frac{7}{20}$. The corresponding picture is analogous. To summarize this example, the process of
bradley_spring22_F00090	17	■ 1.000	$\left(\frac{1}{5}, \frac{1}{4}, \frac{1}{20}, \frac{3}{20}, \frac{7}{20}, \frac{3}{20}, \frac{7}{20}\right)$	$\left(\frac{1}{5}, \frac{1}{4}, \frac{1}{20}, \frac{3}{20}, \frac{7}{20}\right)$	probability distribution is equal to $\left(\frac{1}{5}, \frac{1}{4}, \frac{1}{20}, \frac{3}{20}, \frac{7}{20}\right)$. It is a point in Δ_5 , and its picture is shown
bradley_spring22_F00091	17	■ 1.000	Δ_5	Δ_5	probability distribution is equal to $\left(\frac{1}{5}, \frac{1}{4}, \frac{1}{20}, \frac{3}{20}, \frac{7}{20}\right)$. It is a point in Δ_5 , and its picture is shown
bradley_spring22_F00092	18	■ 0.999	$p \times (q, r)$	$p \times (q, r)$	denote this new distribution by, say, “ $p \times (q, r)$ ” to remind us that the probabilities in q and
bradley_spring22_F00093	20	■ 1.000	$f: \mathbb{R}^2 \rightarrow \mathbb{R}$	$f: \mathbb{R}^2 \rightarrow \mathbb{R}$	of the inputs for a different function. Multiplication, for instance, is a function $f: \mathbb{R}^2 \rightarrow \mathbb{R}$
bradley_spring22_F00094	20	■ 0.856	(x, y)	(x, y)	that accepts a pair of numbers (x, y) as input and computes their product $f(x, y) = xy$ as
bradley_spring22_F00095	20	■ 0.856	$f(x, y)=x y$	$f(x, y) = xy$	that accepts a pair of numbers (x, y) as input and computes their product $f(x, y) = xy$ as
bradley_spring22_F00096	20	■ 1.000	$p=$	$p =$	an operation? It is not, really. It would be better to represent a probability distribution $p =$
bradley_spring22_F00097	20	■ 0.999	$f: \mathbb{R}^n \rightarrow \mathbb{R}$	$f: \mathbb{R}^n \rightarrow \mathbb{R}$	$(p_1, p_2, \dots, p_n)$ by an actual (continuous) function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ that accepts a list of numbers
bradley_spring22_F00098	20	■ 1.000	$x=\left(x_{\{1\}}, x_{\{2\}}, \ldots, x_{\{n\}}\right)$	$x = (x_1, x_2, \dots, x_n)$	$\vec{x} = (x_1, x_2, \dots, x_n)$ as input and that outputs a single number $f(x)$. In such a passage from
bradley_spring22_F00099	20	■ 1.000	$p_{\{1\}}, p_{\{2\}}, \ldots, p_{\{n\}}$	$p_1, p_2, \dots, p_n$	<i>probability distributions to functions</i> , the formula for f may involve the probabilities $p_1, p_2, \dots, p_n$
bradley_spring22_F00100	20	■ 1.000	$x=$	$x =$	somehow. As an example, given a probability distribution p and a list of arbitrary numbers $x =$
bradley_spring22_F00101	20	■ 0.995	$\left(x_{\{1\}}, x_{\{2\}}, \ldots, x_{\{n\}}\right)$	$(x_1, x_2, \dots, x_n)$	$(x_1, x_2, \dots, x_n)$ as input, we could define $f(x)$ to be equal to the sum $p_1x_1 + p_2x_2 + \dots + p_nx_n$, a
bradley_spring22_F00102	20	■ 0.995	$p_{\{1\}} x_{\{1\}}+p_{\{2\}} x_{\{2\}}+\cdots+p_{\{n\}} x_{\{n\}}$	$p_1x_1 + p_2x_2 + \dots + p_nx_n$	$(x_1, x_2, \dots, x_n)$ as input, we could define $f(x)$ to be equal to the sum $p_1x_1 + p_2x_2 + \dots + p_nx_n$, a
bradley_spring22_F00103	21	■ 1.000	p, q	p, q	p, q and r to obtain $p \circ (q, r)$ in our coin-food example is neither homeless nor isolated in the
bradley_spring22_F00104	21	■ 1.000	$p \circ (q, r)$	$p \circ (q, r)$	p, q and r to obtain $p \circ (q, r)$ in our coin-food example is neither homeless nor isolated in the
bradley_spring22_F00105	21	■ 1.000	$H(p \circ (q, r))$	$H(p \circ (q, r))$	tions. In other words, can the formula for $H(p \circ (q, r))$ be reexpressed in terms of $H(p)$ and

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bradley_spring22_F00106	21	■ 1.000	$H(q)$	$H(q)$	$H(q)$ and $H(r)$? Perhaps, for instance, the entropy of the composite distribution is equal to
bradley_spring22_F00107	21	■ 1.000	$H(r)$	$H(r)$	$H(q)$ and $H(r)$? Perhaps, for instance, the entropy of the composite distribution is equal to
bradley_spring22_F00108	21	■ 1.000	$H(p \circ (q, r)) \stackrel{?}{=} H(p) + H(q) + H(r)$	$H(p \circ (q, r)) \stackrel{?}{=} H(p) + H(q) + H(r)$	the sum of the entropies of the individual distributions: $H(p \circ (q, r)) \stackrel{?}{=} H(p) + H(q) + H(r)$.
bradley_spring22_F00109	22	■ 1.000	$q^1, q^2, \dots, q^n$	$q^1, q^2, \dots, q^n$	$q^1, q^2, \dots, q^n$. (Pay careful attention to the difference between subscripts and superscripts:
bradley_spring22_F00110	22	■ 1.000	p_1	p_1	p_1 is a number, whereas q^1 is a list of numbers.) As in our motivating example above, the
bradley_spring22_F00111	22	■ 1.000	q^1	q^1	p_1 is a number, whereas q^1 is a list of numbers.) As in our motivating example above, the
bradley_spring22_F00112	22	■ 1.000	q^2	q^2	q^1 might be an element of Δ_3 , while q^2 might be an element of Δ_{17} , while q^3 might be an
bradley_spring22_F00113	22	■ 1.000	Δ_{17}	Δ_{17}	q^1 might be an element of Δ_3 , while q^2 might be an element of Δ_{17} , while q^3 might be an
bradley_spring22_F00114	22	■ 1.000	q^3	q^3	q^1 might be an element of Δ_3 , while q^2 might be an element of Δ_{17} , while q^3 might be an
bradley_spring22_F00115	22	■ 1.000	$p \circ (q^1, q^2, \dots, q^n)$	$p \circ (q^1, q^2, \dots, q^n)$	distribution $p \circ (q^1, q^2, \dots, q^n)$ satisfies the following important equation, which is sometimes
bradley_spring22_F00116	22	■ 1.000	$\left\{F: \Delta_n \rightarrow \mathbb{R}\right\}$	$\{F: \Delta_n \rightarrow$	► <i>In more detail.</i> This characterization may be stated more formally as follows: if $\{F: \Delta_n \rightarrow$
bradley_spring22_F00117	22	■ 1.000	$\mathbb{R}\}$	$\mathbb{R}\}$	$\mathbb{R}\}$ is any other collection of continuous functions satisfying Equation (5), then F must in
bradley_spring22_F00118	22	■ 1.000	F	F	$\mathbb{R}\}$ is any other collection of continuous functions satisfying Equation (5), then F must in
bradley_spring22_F00119	22	■ 1.000	$F=cH$	$F=cH$	fact be <i>equal</i> to H or to some multiple of it. In symbols this means $F=cH$ for some real
bradley_spring22_F00120	22	■ 1.000	c	c	number c . As mentioned above, this important theorem appears in [Lei21] and is actually a
bradley_spring22_F00121	23	■ 1.000	$\left\{F: \Delta_n \rightarrow \mathbb{R}\right\}_{n \geq 1}$	$\{F: \Delta_n \rightarrow \mathbb{R}\}_{n \geq 1}$	Theorem 1. Let $\{F: \Delta_n \rightarrow \mathbb{R}\}_{n \geq 1}$ be a sequence of functions. The following are equivalent:
bradley_spring22_F00122	23	■ 1.000	$H(p \circ q)=H(p)+H(q)$	$H(p \circ q)=H(p)+H(q)$	form $H(p \circ q)=H(p)+H(q)$ for probability distributions p and q . In words, this equation
bradley_spring22_F00123	23	■ 0.998	$\circ$	$\circ$	seems ⁵ to say that entropy H behaves nicely with composition $\circ$ of probability distributions
bradley_spring22_F00124	23	■ 0.641	$f(\bullet \bullet)=f(\bullet) f(\bullet)$	$f(\bullet \bullet)=f(\bullet) f(\bullet)$	Y ; that is, if the function satisfies the equation $f(\bullet \bullet)=f(\bullet) f(\bullet)$ for all elements $\bullet$ and $\bullet$ in X .
bradley_spring22_F00125	23	■ 0.641	$\bullet$	$\bullet$	Y ; that is, if the function satisfies the equation $f(\bullet \bullet)=f(\bullet) f(\bullet)$ for all elements $\bullet$ and $\bullet$ in X .
bradley_spring22_F00126	23	■ 1.000	$p \circ q=q$	$p \circ q=q$	$p \circ q=q$ and $H(p)=0$ and so the chain rule reduces to $H(q)=H(q)$, which is uninteresting. But we will momentarily
bradley_spring22_F00127	23	■ 1.000	$H(p)=0$	$H(p)=0$	$p \circ q=q$ and $H(p)=0$ and so the chain rule reduces to $H(q)=H(q)$, which is uninteresting. But we will momentarily
bradley_spring22_F00128	23	■ 1.000	$H(q)=H(q)$	$H(q)=H(q)$	$p \circ q=q$ and $H(p)=0$ and so the chain rule reduces to $H(q)=H(q)$, which is uninteresting. But we will momentarily
bradley_spring22_F00129	24	■ 1.000	$q^1=q$	$q^1=q$	⁶ Notice this equation coincides with Equation (4) when $p=\left(\frac{1}{2}, \frac{1}{2}\right)$ corresponds to a coin toss and when $q^1=q$ and
bradley_spring22_F00130	24	■ 1.000	$q^2=r$	$q^2=r$	$q^2=r$ correspond to our breakfast and dinner choices in the example in the previous section.
bradley_spring22_F00131	25	■ 1.000	$f: \mathbb{R} \rightarrow \mathbb{R}$	$f: \mathbb{R} \rightarrow \mathbb{R}$	functions. To elaborate, the derivative of a differentiable function $f: \mathbb{R} \rightarrow \mathbb{R}$ (that is, a function
bradley_spring22_F00132	25	■ 1.000	$f^{\prime}$	$f^{\prime}$	whose derivative exists) is another function often denoted by $f^{\prime}$ or by $\frac{d f}{d x}$ or by $d(f)$. Given
bradley_spring22_F00133	25	■ 1.000	$\frac{d f}{d x}$	$\frac{d f}{d x}$	whose derivative exists) is another function often denoted by $f^{\prime}$ or by $\frac{d f}{d x}$ or by $d(f)$. Given

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bradley_spring22_F00134	25	■ 1.000	$d(f)$	$d(f)$	whose derivative exists) is another function often denoted by f' or by $\frac{df}{dx}$ or by $d(f)$. Given
bradley_spring22_F00135	25	■ 1.000	g	g	two differentiable functions f and g , it is natural to ask about the derivative of their product.
bradley_spring22_F00136	25	■ 1.000	$(fg)^{\prime}$	$(fg)^{\prime}$	Can the derivative $(fg)^{\prime}$ be expressed in terms of the individual functions $f^{\prime}$ and $g^{\prime}$? The
bradley_spring22_F00137	25	■ 1.000	$g^{\prime}$	$g^{\prime}$	Can the derivative $(fg)^{\prime}$ be expressed in terms of the individual functions $f^{\prime}$ and $g^{\prime}$? The
bradley_spring22_F00138	25	■ 0.999	$(fg)^{\prime}(x)=f^{\prime}(x)g(x)+f(x)g^{\prime}(x)$	$(fg)^{\prime}(x)=f^{\prime}(x)g(x)+f(x)g^{\prime}(x)$	the function whose value at a point x is given by $\boxed{(fg)^{\prime}(x)=f^{\prime}(x)g(x)+f(x)g^{\prime}(x)}$. In words,
bradley_spring22_F00139	25	■ 0.978	$f\,g$	fg	the derivative of fg is obtained by multiplying g by the derivative of f , then multiplying f by
bradley_spring22_F00140	25	■ 1.000	$d(f\,g)=d(f)\,g+f\,d(g)$	$d(fg)=d(f)g+fd(g)$	succinctly as $\boxed{d(fg)=d(f)g+fd(g)}$, which should be compared with Equation (8).
bradley_spring22_F00141	25	■ 1.000	d	d	once again intentional. Then, informally speaking, a derivation is defined to be any function d
bradley_spring22_F00142	25	■ 0.322	$d(\bullet\bullet)=d(\bullet)\,\bullet+\bullet\,d(\bullet)$	$d(\bullet\bullet)=d(\bullet)\bullet+\bullet d(\bullet)$	on this set that satisfies the Leibniz rule $d(\bullet\bullet)=d(\bullet)\bullet+\bullet d(\bullet)$ for all elements $\bullet$ and $\bullet$. As one
bradley_spring22_F00143	25	■ 1.000	A	A	makes sense. Given an algebra A , for instance, a derivation on A is formally defined to be a
bradley_spring22_F00144	25	■ 0.737	$[0,1]$	$[0,1]$	? As an example, here is a simple exercise. Let $[0,1]$ denote the set of all numbers between and including 0 and 1, and
bradley_spring22_F00145	25	■ 1.000	$d:[0,1]\rightarrow\mathbb{R}$	$d:[0,1]\rightarrow\mathbb{R}$	define a function $d:[0,1]\rightarrow\mathbb{R}$ by declaring $d(x)=-x\log(x)$ if $x>0$ and $d(x)=0$ if $x=0$. Show that d satisfies
bradley_spring22_F00146	25	■ 1.000	$d(x)=-x\,\log\,(x)$	$d(x)=-x\log(x)$	define a function $d:[0,1]\rightarrow\mathbb{R}$ by declaring $d(x)=-x\log(x)$ if $x>0$ and $d(x)=0$ if $x=0$. Show that d satisfies
bradley_spring22_F00147	25	■ 1.000	$x>0$	$x>0$	define a function $d:[0,1]\rightarrow\mathbb{R}$ by declaring $d(x)=-x\log(x)$ if $x>0$ and $d(x)=0$ if $x=0$. Show that d satisfies
bradley_spring22_F00148	25	■ 1.000	$d(x)=0$	$d(x)=0$	define a function $d:[0,1]\rightarrow\mathbb{R}$ by declaring $d(x)=-x\log(x)$ if $x>0$ and $d(x)=0$ if $x=0$. Show that d satisfies
bradley_spring22_F00149	25	■ 1.000	$x=0$	$x=0$	define a function $d:[0,1]\rightarrow\mathbb{R}$ by declaring $d(x)=-x\log(x)$ if $x>0$ and $d(x)=0$ if $x=0$. Show that d satisfies
bradley_spring22_F00150	25	■ 0.958	$d(x\,y)=d(x)\,y+x\,d(y)$	$d(xy)=d(x)y+xd(y)$	the Leibniz rule. That is, show that $d(xy)=d(x)y+xd(y)$ for all numbers x and y in $[0,1]$. This computation only
bradley_spring22_F00151	26	■ 1.000	$d:A\rightarrow A$	$d:A\rightarrow A$	linear function $d:A\rightarrow A$ satisfying the Leibniz rule $d(ab)=d(a)b+ad(b)$ for all elements
bradley_spring22_F00152	26	■ 1.000	$d(a\,b)=d(a)\,b+a\,d(b)$	$d(ab)=d(a)b+ad(b)$	linear function $d:A\rightarrow A$ satisfying the Leibniz rule $d(ab)=d(a)b+ad(b)$ for all elements
bradley_spring22_F00153	26	■ 1.000	a	a	a and b in A . This can be generalized slightly by replacing the target A with another object
bradley_spring22_F00154	26	■ 1.000	b	b	a and b in A . This can be generalized slightly by replacing the target A with another object
bradley_spring22_F00155	26	■ 0.996	M	M	M called a “bimodule over A ” and instead considering functions $d:A\rightarrow M$ satisfying the
bradley_spring22_F00156	26	■ 0.996	$A^{\prime\prime}$	$A^{\prime\prime}$	M called a “bimodule over A ” and instead considering functions $d:A\rightarrow M$ satisfying the
bradley_spring22_F00157	26	■ 0.996	$d:A\rightarrow M$	$d:A\rightarrow M$	M called a “bimodule over A ” and instead considering functions $d:A\rightarrow M$ satisfying the
bradley_spring22_F00158	26	■ 1.000	$d(a)$	$d(a)$	same equation. These, too, are called derivations. The only difference now is that $d(a)$ is an
bradley_spring22_F00159	26	■ 1.000	$d(a)\,b$	$d(a)b$	of the expressions $d(a)b$ and $ad(b)$. Here, $d(a)$ and b are elements of different sets, namely
bradley_spring22_F00160	26	■ 1.000	$a\,d(b)$	$ad(b)$	of the expressions $d(a)b$ and $ad(b)$. Here, $d(a)$ and b are elements of different sets, namely
bradley_spring22_F00161	26	■ 1.000	$0.5\,\times 8=4$	$0.5\times 8=4$	members of the same set. By way of analogy, multiplication of two numbers 0.5×4 makes
bradley_spring22_F00162	26	■ 0.960	$0.5\,\times$	$0.5\times$	sense, but what would it mean to multiply a number with an English word, $0.5\times yellow=?$

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bradley_spring22_F00163	26	■ 1.000	$d(b)$	$d(b)$	and $d(b)$. Bimodules are the answer to such questions. A bimodule over A is a mathematical
bradley_spring22_F00164	26	■ 0.848	$\mathbf{v}$	$\mathbf{v}$	multiplication. The ability to multiply a vector $\mathbf{v}$ by a real number k to obtain a new vector
bradley_spring22_F00165	26	■ 0.848	k	k	multiplication. The ability to multiply a vector $\mathbf{v}$ by a real number k to obtain a new vector
bradley_spring22_F00166	27	■ 1.000	$\left\{H: \Delta_n \rightarrow \mathbb{R}\right\}$	$\{H: \Delta_n \rightarrow \mathbb{R}\}$	of infinitely many continuous functions $\{H: \Delta_n \rightarrow \mathbb{R}\}$ satisfying a certain equation—the chain
bradley_spring22_F00167	27	■ 0.947	$\left\{d: \Delta_n \rightarrow \boldsymbol{-}\right\}$	$\{d: \Delta_n \rightarrow -\}$	finitely many continuous functions $\{d: \Delta_n \rightarrow \blacksquare\}$ satisfying a certain equation—the Leibniz
bradley_spring22_F00168	27	■ 1.000	$d(p): \mathbb{R}^n \rightarrow \mathbb{R}$	$d(p): \mathbb{R}^n \rightarrow \mathbb{R}$	operad of topological simplices assigns a <i>continuous function</i> $d(p): \mathbb{R}^n \rightarrow \mathbb{R}$ to each probability
bradley_spring22_F00169	27	■ 1.000	$\mathbb{R}^n$	$\mathbb{R}^n$	black box above. It is precisely the set of all continuous functions from $\mathbb{R}^n$ to $\mathbb{R}$. This set can
bradley_spring22_F00170	27	■ 0.921	$\operatorname{hom}(\mathbb{R}^n, \mathbb{R})$	$\operatorname{hom}(\mathbb{R}^n, \mathbb{R})$	Even so, let $\operatorname{hom}(\mathbb{R}^n, \mathbb{R})$ denote this topological space of functions. Then we can summarize
bradley_spring22_F00171	27	■ 1.000	$\left\{d: \Delta_n \rightarrow \operatorname{hom}(\mathbb{R}^n, \mathbb{R})\right\}$	$\{d: \Delta_n \rightarrow \operatorname{hom}(\mathbb{R}^n, \mathbb{R})\}$	<i>continuous functions</i> $\{d: \Delta_n \rightarrow \operatorname{hom}(\mathbb{R}^n, \mathbb{R})\}$ that satisfies an appropriate version of the Leibniz rule,
bradley_spring22_F00172	27	■ 1.000	$d(p \circ q) = d(p) \circ q + q \circ d(q)$	$d(p \circ q) = d(p) \circ q + q \circ d(q)$	“ $d(p \circ q) = d(p) \circ q + q \circ d(q)$ ” for any probability distributions p and q .
bradley_spring22_F00173	27	■ 1.000	$p \circ q$	$p \circ q$	The expression $p \circ q$ does not make much sense, for instance. So far we have only used the
bradley_spring22_F00174	27	■ 1.000	$d(p) \circ q$	$d(p) \circ q$	also need to make sense of the expressions $d(p) \circ q$ and $q \circ d(q)$ that appear in the formula.
bradley_spring22_F00175	27	■ 1.000	$q \circ d(q)$	$q \circ d(q)$	also need to make sense of the expressions $d(p) \circ q$ and $q \circ d(q)$ that appear in the formula.
bradley_spring22_F00176	27	■ 1.000	$d(q)$	$d(q)$	probability distribution, whereas $d(q)$ is a function. What does it mean to combine the two?
bradley_spring22_F00177	27	■ 1.000	$\mathbb{R}^n \rightarrow \mathbb{R}$	$\mathbb{R}^n \rightarrow \mathbb{R}$	⁸ Formally speaking, we can equip the set of continuous functions $\mathbb{R}^n \rightarrow \mathbb{R}$ with what is known as the “product
bradley_spring22_F00178	28	■ 1.000	$d(p \circ (q^1, q^2, \dots, q^n))$	$d(p \circ (q^1, q^2, \dots, q^n))$	the chain rule essentially says that the function $d(p \circ (q^1, q^2, \dots, q^n))$ is obtained by applying
bradley_spring22_F00179	29	■ 1.000	$d(p)(x) = H(p)$	$d(p)(x) = H(p)$	the function $d(p): \mathbb{R}^n \rightarrow \mathbb{R}$ to be constant at entropy; that is, define $d(p)(x) = H(p)$ for all
bradley_spring22_F00180	29	■ 1.000	$d(p)(x) = c H(p)$	$d(p)(x) = c H(p)$	exists a point x in $\mathbb{R}^n$ so that $d(p)(x) = c H(p)$ for some real number c . The proof of this part
bradley_spring22_F00181	29	■ 1.000	$d(p)(0) = c H(p)$	$d(p)(0) = c H(p)$	easily show that the special point x is actually zero, that is, $d(p)(0) = c H(p)$.